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Glasses and Glass-Ceramics as Sealing Materials

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Abstract

The term sealing glass, more than a composition or specific property, describes the use of glass to create a hermetic seal. Glasses are known from antiquity but the use of glass with the purpose of obtaining hermetic seals is relatively recent and its development is linked to the growth of the electronics and semiconductor industry.

The joining of glass/metal or glass/ceramic interfaces is a field of research which covers energy-generation systems, electricity, microelectronics and protection against corrosion. The study of sealing glasses implies an interdisciplinary approach to research and development connecting different areas of knowledge such as the design of glass compositions, melting and processing, adherence and inter-diffusion processes and also the most modern techniques for the study of surfaces and interfacial reactions. This article will cover all these aspects including some relevant compositions, the steps of the sealing process as well as conventional and new applications for glass and glass-ceramic as sealing materials.

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